Alan Khakimov

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Professional Summary

Biomedical Engineering student specializing in Machine Learning and Clinical Software Systems. Experienced in leading the end-to-end development of AI-driven platforms, with a focus on translating complex data pipelines into intuitive, user-centric solutions. Committed to Responsible AI and ethical data frameworks.

Education

University of British Columbia — **Vancouver, BC**
B.A.Sc. in Biomedical Engineering – Software and Machine Learning Specialization — *Sep. 2021 – May 2026*

- **Cumulative GPA:** 3.70/4.33, Dean's List
- **Coursework:** Machine Learning, Algorithms and Data Structures, Statistics, Relational Databases

Experience

Torus Biomedical Solutions — **Vancouver, BC**
Medical Image Analysis Intern, Research and Development — *Sep. 2024 – Apr. 2025*

- Developed and verified training data for an AI-driven intraoperative guidance system for total hip arthroplasty
- Implemented Python-based algorithms for computing personalized pelvic and hip-joint coordinate systems from medical imaging data
- Designed simulation workflows for virtual surgical tool placement across multiple stages of surgery
- Built a database with automated quality assessment workflows, reducing manual review time by 90%
- Designed 50+ hip joint replacement prostheses and surgical tools in CAD

Selected Projects

Clinical Decision Support System – Kidney Transplant Outcomes — *Sep. 2025 – Apr. 2026*

- Leading development of an end-to-end ML pipeline for clinical risk and survival trajectory prediction using dynamic patient data, including cross-validation and performance evaluation of survival models
- Collaborating with Vancouver Coastal Health to design a clinical decision support platform integrating longitudinal patient data, predictive modeling, and clinician-facing interfaces
- Performed data preprocessing, time-to-event handling, feature engineering, and missing-value imputation
- Developed a full-stack application enabling secure upload, visualization, and querying of patient lab and clinical data to support model-driven decision making

Martial Arts Tracking and Video Analysis – Personal Project — *Dec. 2025 – Ongoing*

- Designed and implemented a modular computer vision system for fighter tracking in dynamic training footage
- Developed a computer vision module using YOLO-based person detection and pose estimation for frame-to-frame fighter identification and positional tracking
- Built in a robust state management system, supporting manual overrides, template-based tracking, and redundant safeguards to ensure consistent identity assignment under camera panning, occlusion and ambiguity

Leadership Involvement

Investa Insights — **Vancouver, BC**
Senior Quantitative Analyst, UBC Chapter — *Aug. 2025 – Apr. 2026*

- Selected as one of five senior analysts at one of Canada's largest intercollegiate investment clubs, collaborating with over a hundred students nationwide to research global financial markets
- Co-founded a pilot quantitative research program to apply statistical and machine learning methods in equity research, tailoring to biotechnology and pharmaceutical companies
- Designed and deployed models for NLP, Monte Carlo simulations, factor-regression and risk management
- Mentored junior analysts on Python, ML model design and data analysis fundamentals, as well as promoting cross-collaboration between quantitative and fundamental teams

Technical Skills

Languages: Python, SQL, C/C++, R, MATLAB, Java
Frameworks / Libraries: PyTorch, OpenCV, YOLO, NumPy, Pandas, PyVista, Matplotlib
Developer Tools: Git, Bash, Conda
Certifications: TCPS 2 (CORE) - Human Research Ethics
Athletics: Powerlifting (Pursuing Canada's National Team), Wrestling (Provincial Champion), Baseball
Interests: Entrepreneurship, Strategy, Chess, Cooking